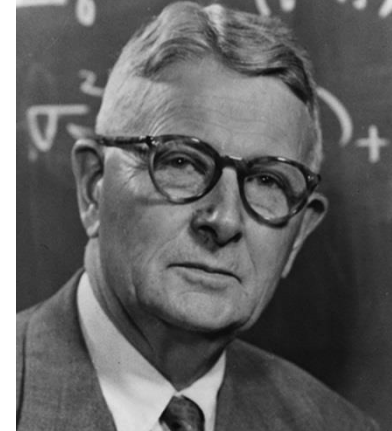
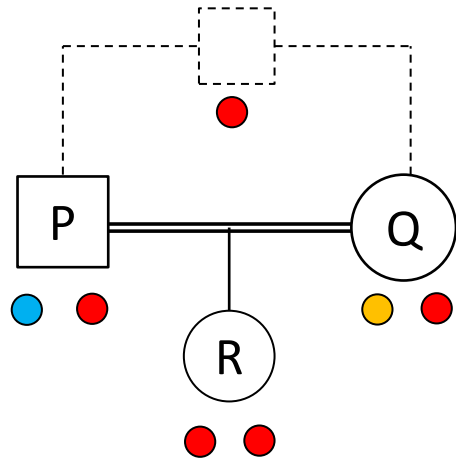


Coefficient of kinship/inbreeding



Sewall Wright
(1889 - 1988)

- Wright (1921): The kinship coefficient φ between P and Q

$$\varphi_{P,Q} = P(\text{random allele of P is IBD with random allele of Q})$$

$$= P(\text{R receive IBD alleles from her parents})$$

$$= P(\text{R is autozygous})$$

$$= f_R$$

the inbreeding coefficient of R

P and Q related



$$\varphi_{P,Q} > 0$$